

The Role of Human-Cat Interactions in Shaping Domestic Cat (*Felis catus*) Hunting Behavior

By

HEE JIN CHUNG  
DISSERTATION

Submitted in partial satisfaction of the requirements for the degree of

DOCTOR OF PHILOSOPHY

in

Animal Behavior

in the

OFFICE OF GRADUATE STUDIES

of the

UNIVERSITY OF CALIFORNIA

DAVIS

Approved:

---

Andrew Sih, Chair

---

Carly Moody, Co-Chair

---

Benjamin N. Sacks

Committee in Charge

2026

## Table of Contents

|                                                                                                                                                             |            |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| <i>Abstract</i> .....                                                                                                                                       | <i>III</i> |
| <i>Acknowledgements</i> .....                                                                                                                               | <i>VI</i>  |
| <i>Dedication</i> .....                                                                                                                                     | <i>IX</i>  |
| <i>Chapter 1: Safe or sound? Factors influencing outdoor access, cat behavior, and hunting history with implications for conservation and welfare</i> ..... | <i>1</i>   |
| ABSTRACT .....                                                                                                                                              | 1          |
| INTRODUCTION .....                                                                                                                                          | 3          |
| METHODS .....                                                                                                                                               | 6          |
| RESULTS .....                                                                                                                                               | 11         |
| DISCUSSION .....                                                                                                                                            | 21         |
| CONCLUSION .....                                                                                                                                            | 27         |
| ACKNOWLEDGEMENTS .....                                                                                                                                      | 28         |
| REFERENCES .....                                                                                                                                            | 28         |
| <i>Chapter 2: Utilizing citizen science to assess consistent individual variation in play behavior of the domestic cat, Felis catus</i> .....               | <i>33</i>  |
| ABSTRACT .....                                                                                                                                              | 33         |
| INTRODUCTION .....                                                                                                                                          | 34         |
| METHODS .....                                                                                                                                               | 36         |
| RESULTS .....                                                                                                                                               | 44         |
| DISCUSSION .....                                                                                                                                            | 51         |
| CONCLUSION .....                                                                                                                                            | 55         |
| ACKNOWLEDGEMENTS .....                                                                                                                                      | 56         |
| REFERENCES .....                                                                                                                                            | 57         |
| <i>Chapter 3: Using DNA metabarcoding and citizen science to assess the diet of owned domestic cats</i> .....                                               | <i>60</i>  |
| ABSTRACT .....                                                                                                                                              | 60         |
| INTRODUCTION .....                                                                                                                                          | 61         |
| METHODS .....                                                                                                                                               | 64         |
| RESULTS .....                                                                                                                                               | 69         |
| DISCUSSION .....                                                                                                                                            | 76         |
| CONCLUSION .....                                                                                                                                            | 80         |
| REFERENCES .....                                                                                                                                            | 81         |

## ABSTRACT

Domestic cats, *Felis catus*, play an interesting role in human society, both as companion animals and as predators of local wildlife. Cats are one of the most commonly kept companion animals, surpassing dogs in many areas around the world, and are frequently directly managed by caretakers. However, cats are also one of the most widespread mesopredators with the ability to impact vulnerable prey species, and when outdoors (both owned and unowned) can cause prey population decline or even extinction. This has led to much discussion among scientists and the public on the proper management of owned cats, and whether possible benefits for their welfare warrant the provisioning of free-roaming outdoor access. The objective of this dissertation was to assess the role of management decisions and human-cat interactions on cats' hunting behavior, bringing together the perspectives of behavioral ecology, conservation ecology, animal welfare, and anthrozoology.

Many efforts have been made to reduce cats' hunting of wildlife, spanning from population control measures such as lethal population control and Trap-Neuter-Release, to targeting caretaker-led interventions such as reduced outdoor access and increased interactive play in the home. Some methods designed to reduce cat impact on wildlife can raise concern among caretakers, particularly if they are perceived to negatively impact their cat's welfare. This research focused on understanding the role of caretakers, not only as a mediator of cat-wildlife interactions, but also as partners of data collection on their cats' hunting and hunting-related behaviors. We relied on citizen science to better understand the perspectives and motivations of cat caretakers regarding outdoor access and hunting (Chapter 1), quantify individual variation in cats' hunting-related behaviors (Chapter 2), and test molecular methods of scat analysis for detection of prey consumption (Chapter 3).

In Chapter 1, we aimed to explore attitudes towards providing or reducing outdoor access, identify specific predictors of free-roaming outdoor access, and identify predictors of hunting behavior among cats that are allowed to freely roam. Using a quantitative questionnaire (N=361), we found that more than half of outdoor cat caretakers were likely or somewhat likely to reduce outdoor access if they lived near busier streets (59%), if more dangerous animals were around (59%), if their cat stopped coming home consistently (56%), or if they knew their cat was negatively impacting wildlife (53%). We found that cats had higher odds of freely roaming if they were older, obtained from the street, living close to traffic, or if they showed aggressive behaviors in the home. Among those allowed outside, cats were more likely to have hunted prey in the last year if they were allowed to freely roam instead of being supervised, or were allowed outdoors between in the morning hours.

In Chapter 2, we recruited participants in the United States (N=87) to expose their cat to standardised behavioral assay trials using a wand toy, to measure individual variation in response to the toy and the consistency of their responses over time. We found high repeatability of all behaviors indicative of cat interest measured across trials (mean  $R = 0.467$ ). We then calculated a composite interest score consisting of all behavior variables, and found significant consistent individual variation in the cats' interest in the toy (ICC = 0.717). Interestingly, we did not find any significant connections between the cats' interest score and other factors of interest relating to their management in the home. This study was the first recorded use of citizen science to quantify consistent individual differences in play behavior of cats, and highlights the importance of assessing individual variation.

In Chapter 3, we relied on owner-collected scats ( $n = 32$ ) from litterboxes and backyards to test whether DNA metabarcoding could be used to detect consumption of wild prey by owned,

indoor-outdoor domestic cats. We found only one instance of wild prey consumption, a California vole (*Microtus californicus*), despite the caretaker not witnessing any prey returns during the 24-hour period prior to scat sampling. Conversely, in the instances where the caretaker reported returns of intact prey by their cats, prey DNA was not detected in their scat. This study presents the first recorded use of DNA metabarcoding to detect consumption of wild prey by owned domestic cats, and suggests that caretaker reports and scat analysis should be used in combination to obtain a more thorough understanding of their hunting of wild prey. Together, this dissertation research provides much needed insight into the role of cat caretakers in mediating cat-wildlife dynamics, such as through their influence on their cats' outdoor access and behavior, and highlights their potential and importance as partners of data collection when studying companion animal behavior.

## Acknowledgements

Words are not sufficient to show my appreciation for the wonderful mentors who have supported me throughout my dissertation. I would first like to thank Andy for believing in me coming into graduate school. Andy has taught me how to be a broad thinker, to host karaoke parties, and how to cultivate an intellectual community that fosters curiosity and critical thinking. Thank you for your generosity and patience as I navigated many challenges and milestones throughout this degree.

I would like to thank Carly for taking me on as a graduate student even though I knew next to nothing about cats. Carly has taught me how to not miss the details, to maintain academic rigor, and how to communicate my science to broad audiences. Thank you for your unwavering support, for your generosity with your time and resources, and always making sure my well-being came first.

I would like to thank Ben for welcoming me into his lab, which has been such a valuable community I have gotten to be a part of for the past few years. Thank you for your genuine care for your students, your honest and constructive feedback on ideas and methodology, and your patience as I tried to distinguish the baby from the bathwater.

I want to extend my gratitude to the labs that I have been a part of. Thank you to the Sih lab for the weekly lab meetings, and for willing to delve deep into some idea or project together, sometimes for 3 hours at a time. Thank you especially to Sonja Wild and Kaz Uyehara for dissecting my projects with me for hours, for lending an ear, and for feeding me high-protein snacks and meals. Thank you to all members of the Animal Welfare Epidemiology lab, for all your cat and welfare knowledge and expertise, for all the countless hours of feedback on surveys and study design, and helping me slowly establish myself as a cat scientist. Thank you especially

to Jen Link for being my lab friend and amazing colleague. I would like to thank all members of the Mammalian Ecology and Conservation Unit, for providing me a home where I felt welcomed, and for letting me listen in and ask basic questions about genetics and genomics when I know very little. Thank you especially to Stevi Vanderzwan for being our wonderful lab manager, for training me on new skills, and for all that you do for the lab.

I would like to thank all of the participants of my studies for making this research possible. Thank you for your dedication to learning more about these wonderful animals. Thank you to all the cats that participated (some more eagerly than others). I would like to express my gratitude towards all cat caretakers, cat behaviorists, and enthusiasts, for helping cats all over the world be better understood and cared for.

None of this would have been possible without the support of my peers, my fellow graduate students that I leaned on for support through various challenges. I want to thank all members of the Animal Behavior Graduate Group for being my community, particularly my cohort, Cassidy Cooper, Isabelle McDonald-Gilmartin, Jacob Johnson, Alice Michel, Sabrina Seaborn, and Kirsten Sheehy for starting graduate school with me in 2020 and going through this together. I feel grateful to have found an amazing group of friends who were there with me through good times and bad, especially Sofia Meléndez Cartagena, Deedee Chao, Francine DeCastro, Jordan Colby, Emmarie Purvis, the past and current Halsey crew, and the Crawlspace crew. Thank you to my best friend Natalie Schweizerhof for predicting that I will be a Dr. Chung someday. Thank you all for being part of my community, and for all the moments that we shared that kept me going.

I would like to thank all of my healthcare providers, who have supported me through multiple surgeries and health issues throughout graduate school and thus made this possible. I

especially would like to thank the UCD Student Health Physical Therapy Department, Jocelyn Lam, and Abby Hollandsworth for helping me through the toughest moments.

I would like to thank Harry Anselmo (Mr. A) for first igniting my passion for science back in 7<sup>th</sup> grade. I would not be here if it weren't for you and your dedication to raising the next generation of scientists and curious thinkers.

Finally, I would like to thank my wife, Elsie Platzer, for her unconditional support and love throughout graduate school, and in our lives together. Thank you for being the best stepmom to our cats, and for being my life partner and constant source of fun and inspiration. None of this would have been possible without you. To Memi and Zoe, for being yourselves, and for being willing to be my test subjects #0.

## Dedication

For all animals around the world, big and small. For the cats in our lives, inspiring curiosity in us by being curious themselves. For Rocket, who made me decide to study cats – I hope you are in a better place now.



# **Chapter 1: Safe or Sound? Factors influencing outdoor access, cat behavior, and hunting history with implications for conservation and welfare**

Published in *Applied Animal Behaviour Science* (2024,

<https://doi.org/10.1016/j.applanim.2024.106425>)

## **ABSTRACT**

Outdoor access may provide mental stimulation and enrichment for companion cats (*Felis catus*); however, it also raises concerns regarding their safety and predation of vulnerable local wildlife. Understanding factors that shape cats' outdoor access and behavioral outcomes is therefore crucial for identifying nuanced solutions to the human-cat-wildlife conflict. In this exploratory study, we aimed to 1) explore owner attitudes towards providing or reducing outdoor access, 2) identify general predictors of free-roaming outdoor access, and 3) identify predictors of hunting behavior among cats that are allowed outside. Using a quantitative questionnaire (n=75 questions), we collected information on cat owners' demographics and attitudes, their cat's characteristics and behaviors in the home, home environment, outdoor lifestyle, and hunting history. Of 631 cat owners living near Davis, California, USA, recruited through convenience and snowball sampling, 36.7 % indicated they were keeping their cat indoor-only (n=232), 34.1 % providing supervised or controlled outdoor access (n=215), and 29.2 % providing uncontrolled, or free-roaming outdoor access (n=184). More than half of outdoor cat owners reported they were likely or somewhat likely to reduce outdoor access if they lived near busier streets (59 %, n=235), if more dangerous animals were around (59 %, n=235), if their cat stopped coming home consistently (56 %, n=223), or if they knew their cat was negatively impacting wildlife (53 %, n=211). We found that cats had higher odds of freely roaming if they were 7–10 years old compared to <1 year old (p=0.030), obtained from the street rather than a

shelter ( $p=0.007$ ), living within 500 ft of traffic ( $p=0.002$ ), living with a dog ( $p=0.014$ ) or a child ( $p=0.005$ ), or if they showed aggressive behaviors in the home ( $p=0.008$ ). Cats had lower odds of freely roaming if they had scratching areas in the home ( $p=0.017$ ), or were living with one ( $p=0.020$ ) or two ( $p=0.017$ ) other cats. Among those allowed outside, cats had higher odds of having hunted prey in the last year if they were allowed to freely roam without supervision ( $p=0.016$ ), or were allowed outdoors between 6:00–11:59 am ( $p=0.014$ ). These results generate directions for future research in the field of companion animal management, and emphasize the role of cat owners in the intersection of animal welfare and wildlife conservation.

## INTRODUCTION

The domestic cat (*Felis catus*) is one of the most common household pets, present in approximately 37 million households in the United States (AVMA, 2022). Despite their long association with humans (Driscoll, Macdonald, and O'Brien 2009) and importance as pets in modern homes, cats retain several traits seen in their ancestor, *Felis silvestris lybica*. Cats exhibit obligate and hyper-carnivory, often supplementing their nutritional need for animal-based protein by predation (Eisert 2011; Cecchetti, Crowley, and McDonald 2021). Even when their nutritional needs are met, they exhibit surplus hunting (Yamaguchi et al. 2004; Bradshaw 2006; 2016), which have been observed in both owned (Lloyd et al. 2013) and unowned (McGregor et al. 2015) domestic cats. Such traits have and continue to make them excellent biocontrol agents against rodents in association with humans (Pongrácz and Lugosi 2024), but the impacts of their surplus hunting on local wildlife have sparked debate among conservation biologists, ecologists, and animal welfare experts regarding best practices for owned cat management (Machado et al. 2021; Foreman-Worsley et al. 2021; Rochlitz 2005).

On one hand, outdoor domestic cats have been implicated in approximately 26% of the world's bird, reptile, and mammal species extinctions (Doherty et al. 2016), and are well documented as effective non-native predators in both rural and urban environments (Baker et al. 2008; Balogh, Ryder, and Marra 2011; Lepczyk et al. 2023; Loss and Marra 2017). In addition, uncontrolled outdoor access can impact cat health and safety through transmission of communicable disease and parasites (Burling et al. 2017; Tan, Stellato, and Niel 2020), trauma (Olsen and Allen, 2001), predation (Timm et al., 2004), or injury or death through encounters with other animals or automobiles (Tan, Stellato, and Niel 2020). Lastly, while not a problem in the United States, in other regions of the world, the risk of hybridization between free-roaming

domestic cats and other related species (e.g., the European wildcat) is of serious conservation concern (Nussberger et al. 2023), especially given that cat owners have been shown to manage their cats irrespective of their knowledge of such risks (Pongrácz et al. 2024).

On the other hand, restriction to indoor-only living without the provision of appropriate resources may compromise cat welfare, potentially leading to behavioral problems such as house-soiling (periuria), excessive activity and vocalization, and aggression towards other animals and humans (Amat et al. 2009; Amat, Camps, and Manteca 2016; Strickler and Shull 2014; Heidenberger 1997). Outdoor access is also associated with physical activity and thus may decrease their risk of obesity, which may have implications for their overall health and well-being (Rowe et al. 2015). The American Association of Feline Practitioners (AAFP) suggests a series of environmental, social, and behavioral recommendations for enhancing indoor companion cat welfare (Ellis et al. 2013), such as hiding areas, separated areas with various resources, and opportunities for play and predatory behavior; however, providing these resources may not always be viable or desirable for owners, which may lead owners to rely on outdoor access as an alternative.

Much focus in the literature has been on quantifying the effect of both owned and unowned outdoor cats on wildlife. Recently, studies have identified the need to include cat-owners in the discourse (Crowley, Cecchetti, and McDonald 2019; 2020), with increased focus on understanding owner attitudes and perspectives in order to implement change in management decisions (McLeod, Hine, and Bengsen 2015; Linklater et al. 2019). Some studies have surveyed cat owners on their attitudes, cat's characteristics, and living environment to predict uncontrolled outdoor access (Clancy, Moore, and Bertone 2003; McLeod, Hine, and Bengsen 2015; Foreman-Worsley et al. 2021; Tan et al. 2021). However, few studies to date also include information

about the cat's hunting behavior (e.g. Escobar-Aguirre et al. 2019), distinguish between supervised and free-roaming outdoor access, or explicitly ask owners if certain hypothetical scenarios would make them reduce outdoor access. We attempt to fill these knowledge gaps and explore the connections between several welfare-focused factors (e.g. behavior problems, presence of necessary resources in the home) as well as conservation-oriented factors (e.g. hunting history, degree and type of outdoor access) in one sample. We examined: 1) factors that may predict provisioning of free-roaming outdoor access, 2) factors that may predict hunting history among cats that are allowed outside, and 3) owners' reported likelihood to alter outdoor access in response to hypothetical scenarios.

Although this was an exploratory study, we had *a priori* predictions about the relationship between some management factors and cat characteristics and the provisioning of outdoor access. Specifically, based on previous studies, we predicted that cats taken in from the street (Clancy, Moore, and Bertone 2003), living with other animals in the home (Clancy, Moore, and Bertone 2003; Tan et al. 2021), or living farther from traffic and closer to green space (Foreman-Worsley et al. 2021) would be more likely to be allowed outside. We also predicted that male cats (Tan et al. 2021; MacDonald, Milfont, and Gavin 2015), and older cats (Foreman-Worsley et al. 2021; Tan et al. 2021) would be more likely to be allowed outside to freely roam. Regarding hunting history, based on results found by Escobar-Aguirre et al. 2019, we predicted that cats would be more likely to have hunted in the last year if they were living closer to green space, or allowed outside to roam freely rather than be supervised. Additionally, given the activity patterns of potential prey species of cats, we predicted that the time of day the cat is allowed outside would be a significant factor influencing their hunting history (Woods, McDonald, and Harris 2003; McDonald et al. 2015; Linklater et al. 2019).

Lastly, given that owners have been shown to be motivated by their cat's safety (Woods, McDonald, and Harris 2003; McDonald et al. 2015; Linklater et al. 2019) but not their cat's effect on wildlife (Van Eeden et al. 2021; MacDonald, Milfont, and Gavin 2015), we predicted that they would report being likely to reduce outdoor access under hypothetical scenarios that threaten their cat's safety, but not scenarios that suggest their cat's increased impact on wildlife.

Given the disjointed and controversial rhetoric surrounding the topic of outdoor cats, especially between the fields of animal welfare and wildlife conservation, we recognize the importance of considering the interests of both fields when tackling this issue. While various elements of our study were individually explored in previous studies, we tackle these elements together in one sample for the first time and emphasize the need for further connection between different fields in the topic of human-cat-wildlife dynamics.

## **METHODS**

### ***Ethical Note***

This study was reviewed and determined exempt (non-human subjects research) by the Institutional Review Board of the University of California, Davis (#1905569). A consent form including the inclusion criteria and a detailed description of the objectives of the study was presented to each participant at the beginning of the questionnaire. No identifiable information was asked of the participant, and each participant was given an option to click on a separate link at the end of the questionnaire to provide their email to enter a raffle for a chance to win one of three \$50 Amazon gift cards. Each participant was also given an option to click on a separate link to provide their email to be contacted for a follow-up study with their cat.

### ***Study Design***

A cross-sectional, quantitative questionnaire was created and distributed online using Qualtrics™ survey software (Provo, Utah, USA). To participate, respondents had to be at least 18 years old, the current primary caretaker of at least one cat, and reside in one of the following cities in Northern California: Davis, Woodland, Dixon, Winters, West Sacramento, or Sacramento. These cities were selected based on their proximity to the main campus of the University of California, Davis, since this questionnaire was partly distributed to recruit participants for a separate in-person study. Participants were not eligible for the study if they were caretakers of outdoor-only cats who never spent time indoors, given we were interested in assessing the interactions between the cats, the owners, and the home environment. Each participant who completed the questionnaire and provided their email was entered into a raffle to win one of three \$50 Amazon gift cards as an incentive to participate in the study.

This questionnaire (N=75 questions) contained questions on the owners' demographics (n=4), their opinions about outdoor cats (n=7), motivations for letting or not letting their cat outside (n=7), cat characteristics and behavior (n=25), cat-human interactions (n=5), the home environment (n=11), and the cat's outdoor lifestyle and hunting-related behaviors (n=17). If the participant had multiple cats, they were directed to select the cat whose name appears first in the alphabet and answer the rest of the questionnaire about that cat. We used the MacArthur Scale of Subjective Social Status (Adler et al. 2000) as a proxy for the respondents' perceived self-status in relation to their community, since subjective social status has been found to be more informative than objective measures of social status (e.g. Präg, Mills, and Wittek 2016); participants were asked to place themselves on a ladder from a scale of 1 to 10, with 1 being the best off in their community and 10 being the worst off in their community. These questions were chosen based on previous studies on this topic and factors for which we had *a priori* hypotheses.

The study minimum sample size ( $n=360$ ) was calculated by using a standard sample size calculation using a 95% confidence interval and 5% margin of error, with the target population of cat owners estimated to be 25.4% (AVMA, 2022) of the source population of each city (Davis, Woodland, Dixon, Winters, West Sacramento, Sacramento, CA).

### ***Data Collection***

The questionnaire was distributed online from August 1st, 2022 to August 11th, 2022 using convenience and snowball sampling. Given our regional source population, an advertisement was distributed to local community and academic listservs and social media groups. Sampling was terminated when we no longer obtained any new responses, with a final number of 1230 responses.

### ***Data Cleanup & Preparation***

Following data collection, 97 incomplete responses were removed. 53 additional responses that did not meet the inclusion criteria were removed. Lastly, responses of outdoor-only cat owners, and responses with duplicate IP addresses were removed, thus leading to a final sample size of  $N=631$ . All ordered and binary categorical variables were converted into numerical values, and these codes were recorded on a separate spreadsheet. Open text box responses associated with each “Other” response selection were removed, and all variables were coded. The cleaned dataset was inputted into R Statistical Software (v4.2.2; R Core Team 2022). All statistical analyses and visualizations were performed in R. Descriptive statistics were run for all variables included in the survey, and frequency and percentage tables were generated for the variables to be included in the models in STATA™ statistical software (College Station, TX: StataCorp, LLC) prior to model building. The distribution of all biologically plausible predictor variable response categories for outdoor access and hunting history were visually assessed, and

sparsely distributed categories ( $n < 20$ ) were collapsed when appropriate in order to increase accuracy of estimation parameters.

### ***Logistic Regression***

Binary logistic regression models were built to assess biologically relevant factors that predicted the following two outcomes: 1) being provided free-roaming outdoor access (yes, no), and 2) having a history of hunting within the past year (yes, no). A large number of predictor variables were involved in modeling the probability of being provided free-roaming outdoor access – thus, in order to avoid overfitting given the sample size, these variables were divided and assessed in two different models: cat characteristics and home environment predictor variables (Model 1), and owner demographics and opinion predictor variables (Model 2). For these models, we compared indoor-only to free-roaming outdoor cats and omitted supervised outdoor cats. This was due to a lack of a clear boundary between supervised outdoor and free-roaming outdoor cats; although owners were given examples of supervised and controlled outdoor access (e.g. catios, leash-walking), the method of supervision or control may heavily influence the cat's degree of interaction with local wildlife.

Cat characteristics and home environment variables of interest (Model 1) included: number of dogs in the household (0, 1, 2+), number of cats in the household (1, 2, 3, 4+), number of children in the household (0, 1, 2+), presence of vertical space (0/1), presence of scratching areas (0/1), number of litter areas (0, 1, 2, 3, 4, 5, 6+) minus the number of cats (1, 2, 3, 4, 5+), the method in which the cat was obtained (animal shelter, street, friend or family, breeder, bred at home, or other), declaw status (0/1), age of cat (<1 year, 1-6 years, 7-10 years, 11+ years), and the current presence or absence of each of the five categories of the most common behavior problems (house-soiling, aggressive behaviors, destructive behaviors, fearfulness, nuisance

behaviors, excessive vocalization, and gastrointestinal and ingestive disorders). We subtracted the number of cats in the home from the number of litter areas due to current guidelines (Ellis et al. 2013) recommending one more litter areas than the number of cats in the household. This calculation allowed us to assess the relationship between the owner's adherence to this recommendation and their likelihood to provide outdoor access.

Owner demographics and opinions variables of interest included: age of the owner (18-24, 25-34, 45-54, 55-64, 65-74, 75+ years old), gender of the owner (man, woman, non-binary, or prefer to self-describe), subjective social status of the owner (self-scoring from 1 to 10, with 1=worst off in the community, and 10=best off in the community), city (Davis, Sacramento, West Sacramento, Dixon, Winters, or Woodland), distance to green space and traffic (<500ft, 500ft – 0.25mi, 0.25mi-0.5mi, 0.5mi-1mi, 1mi-2mi, or 2+ miles), and Likert-scale responses (“strongly disagree”, “somewhat agree”, “neither agree nor disagree”, “somewhat agree”, “strongly agree”) to the following statements: “Some cats prefer to be indoors”; “Some cats prefer to be outdoors”; “All cats should be allowed outside”; “No cat should be allowed outside”; “Cats are positive for the environment”; “Cats are negative for the environment”; and “I am responsible for my cat's effect on wildlife”. Biologically plausible two-way interaction between distance to traffic and response to the statement “All cats should be allowed outside” was tested.

The third logistic regression model was built to examine factors that may plausibly predict the outcome of having a history of hunting within the past year (yes, no), given that the cat is allowed outdoors. The explanatory variables in this base model included: whether or not the cat goes outside during any of the following groups of hours (morning: 6am-11:59am, afternoon: 12pm-5:59pm, evening: 6pm-11:59pm, and overnight: 12am- 12am-5:59am), total number of groups of hours outside, type of outdoor status (free-roaming vs. supervised),

frequency of interactive object play (never, monthly, weekly, or daily), frequency of nail trims (never, once a year, every few months, once a month, every few weeks, or once a week), distance to green space and traffic, age of cat, and sex of cat.

For all models, each predictor variable was tested individually against the outcome variable, and only variables less than or equal to a liberal p-value of 0.2 were retained (Ferguson and Cox 1993). The final models were built following the stepwise selection method with Akaike Information Criterion (AIC) using the `stepAIC()` function in the MASS package in R (Venables and Ripley 2002). Non-significant variables were left in the model if model fit increased with those variables. Variance inflation factors were assessed using the `regclass` R package (v1.6; Petrie 2020). Tukey's adjustment was applied for the p-values and confidence intervals associated with post-hoc pairwise comparisons with four or more categories to reduce the chance of Type 1 errors.

## RESULTS

### *Owner and Cat Demographics*

Participating cat owners (N=631) were mostly women (76.6%) with no children (67.8%) residing in Davis (56.4%) and Sacramento (23.1%), California. The respondents were most frequently between 25 and 34 years old (39.0%). The reported mean Subjective Social Status score was 6.44 (min=1, max=10, 95% CI=3.28, 9.60). Thirty-seven percent of respondents indicated their cat was kept indoor-only, 34.1% indicated providing supervised or controlled outdoor access, and 29.2% indicated providing uncontrolled, or free-roaming outdoor access.

Fifty-one percent of the respondents' cats were male and 48.8% were female. The majority of the cats were neutered (90.2%), and not declawed (87.7%), and most frequently